

AI-Based Food Freshness Detection: Methods and Recommended Strategies for Safe Use

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Abstract- Food spoils fast, especially in the heat and humidity of summer and in the monsoon season. Most of the people rely on checking expiry dates to decide whether food is safe to eat or not, and this guesswork leads to a lot of unnecessary waste of food. This research paper tells us how artificial intelligence, especially computer vision and deep learning models can solve that problem by detecting food freshness from the images. We review the approaches used for this, including convolutional neural networks, transfer learning methods along with their comparison in terms of accuracy, performance and time complexity. We also talk about where this automated strategy feels short - primarily that it can only judge what is visible in an image. It cannot catch the contamination that does not change how food looks. Based on this, we propose a simple framework that combines image processing techniques, CNN based image classification, confidence score, and recipe suggestions for the food that is safe to eat. The takeaway from this literature is encouraging: these models can classify the freshness of food fairly well, but making them work more reliably and accurately depends on better and diverse datasets, lighter models for mobile use and designing the system to be cautious rather than overconfident about the food safety.

Index Terms- food freshness detection, deep learning, convolutional neural networks, computer vision, food waste reduction

I. INTRODUCTION

Food spoilage and wastage are the major global concerns which impacts on environmental sustainability, household expenditure, and food availability. A large amount of food waste occurs at the consumer level, partly because of uncertainty in identifying the freshness and quality of food items manually. Traditional approaches to identifying food quality, such as relying on expiry dates and personal judgment, can be inconsistent. Therefore, developments in artificial intelligence and computer vision have opened new possibilities for automated food freshness detection. Deep learning models such as convolutional neural networks (CNNs) have shown strong effectiveness in image-based classification across agriculture, healthcare, and object recognition. When applied to food analysis, these models can evaluate visible attributes such as texture changes and color variations. This work reviews an AI-based approach that combines image-based food freshness detection with recipe recommendations to support timely food utilization and reduce food waste.

II. DEVELOPMENT METHODOLOGY

This review analyzes existing AI and computer vision techniques for food freshness detection and considers how their outputs can be used to support food-safety decisions. The methodology focuses on image-based food freshness classification using datasets of images with preassigned labels for training a model. The input image is preprocessed, passed through a training model, and analyzed using the prediction and its confidence score. A recommendation module can then suggest ways to use food that appears visually acceptable, while clearly separating recipe support from food-safety confirmation.

A. Data Collection

The first step in an AI-based food freshness detection system is the collection of suitable datasets. Images should represent different types of foods under varied environmental conditions. The dataset can be organized into three main classes: fresh, medium, and rotten. The fresh class represents food that appears suitable for use, the medium class

represents food showing visible signs of deterioration, and the rotten class represents food with clear visual spoilage features. For a reliable model, a diverse dataset is important because models trained on narrow image conditions may perform poorly on unseen images. Recent computer-vision work on fruit and vegetable freshness has also used the three-class fresh, medium, and rotten setting, showing that model performance depends strongly on the dataset and experimental setup [5].

B. Image Preprocessing

Image preprocessing is a vital component for improving consistency. The input image is resized to the dimensions required by the deep learning architecture, and pixel values are normalized to an appropriate range. Computer vision techniques can also be used to improve image quality and remove unnecessary background information.

C. Feature Extraction and Classification

Convolutional neural networks are used as the primary deep learning approach for extracting visual features and classifying food freshness. CNNs automatically learn important characteristics of food, such as color patterns, texture variations, surface damage, and visible spoilage, from input images. A transfer learning approach can also be applied using pretrained architectures such as VGG16, MobileNet, and EfficientNet [1], [3], [4]. In this approach, a pretrained network is used as a feature extractor while the final classification layers are modified for the required freshness classes. The final layer produces scores for each class, and the class with the highest score is selected as the predicted freshness category.

D. Model Training

The whole dataset is divided into training, validation, and testing subsets. The training set is used to learn the parameters of the model, while the validation set is used to monitor performance and adjust hyperparameters. During training, an appropriate loss function such as categorical cross-entropy is minimized using an optimization algorithm such as Adam. Techniques such as early stopping, dropout,

and learning-rate adjustment can also be applied to reduce overfitting and improve generalization.

E. Performance Evaluation

The performance of a food freshness classification model should be evaluated using several metrics rather than accuracy alone. These metrics include accuracy, precision, recall, and F1-score. Accuracy measures the overall percentage of correctly classified images. Precision indicates how accurately a model classifies a particular freshness class, while recall measures its ability to identify all samples belonging to that class. The F1-score provides a balance between precision and recall. A confusion matrix is also useful for analyzing classification errors between fresh, medium, and rotten categories and for identifying which conditions are difficult for the model to distinguish.

F. Confidence Estimation and Decision Making

After classification, the model produces a score associated with the predicted class. When the model produces a low-confidence result, the system can request another image with improved lighting, a closer view, or a different angle. This mechanism can reduce errors caused by poor-quality images. However, a high model score should not be treated as proof that food is safe to consume; it is only an indicator of the model's confidence in the visual classification.

G. Recipe Recommendation

A recommendation module can be integrated into the food freshness classifier to increase its practical usefulness. When food is classified as fresh or visually acceptable, the system can provide suitable recipe recommendations based on the identified food item. This module can use an external recipe API to retrieve recipes related to the detected food. The recipe recommendations should not be interpreted as confirmation that food is safe to consume. They are intended only to help users make timely use of food that appears visually acceptable.

H. Real-World Deployment

A major consideration in practical food freshness detection is deployment on devices such as smartphones. Larger deep learning models may require significant memory, computational resources, and processing time. Therefore, lightweight architectures such as MobileNet are useful candidates for mobile deployment [3].

I. Recommended Strategies for Safe Use

AI-based food freshness detection should be considered a decision-support tool rather than a complete food-safety plan. Visual appearance alone cannot detect all forms of contamination, including pathogens that may not produce visible changes. Food-safety guidance therefore recommends considering storage conditions, handling history, and other relevant information rather than relying only on appearance [8].

The system should clearly communicate uncertainty when the confidence is low and recommend additional inspection when visual evidence indicates severe spoilage. It should not encourage consumption solely because the AI system predicts that food appears fresh. This safety-oriented approach is important for minimizing the risk of false reassurance.

J. Overall System Workflow

The complete workflow consists of the following stages: image dataset collection -> image preprocessing -> CNN-based feature extraction -> freshness classification -> confidence estimation -> decision making -> recipe recommendation.

The input image is first captured or uploaded by the user. It is then preprocessed and passed to the trained deep learning model. The model predicts the freshness category and generates a confidence score. Based on the confidence level and predicted class, the system either provides the classification result, requests another image, or provides appropriate guidance. For suitable food items, the recommendation module can then retrieve relevant recipes.

III. FOOD FRESHNESS DETECTION METHODS

A. Traditional Computer Vision Methods

Traditional computer vision methods can be used to identify visible changes in food that may indicate spoilage. These methods extract color, texture, shape, and surface-related features from food images.

Color analysis examines changes in the color of food. For example, fresh food often has a characteristic color, while spoilage may cause darkening, browning, yellowing, or other visible changes.

Texture analysis evaluates surface patterns and appearance. Changes in food texture, such as surface irregularities or mold-like patterns, can provide useful information about whether food is visually fresh or deteriorated.

Shape and size analysis can also be used to identify changes in food. Some foods may shrink, deform, or change their surface structure as they deteriorate.

Image-based segmentation separates the food region from the background so that relevant foreground features can be analyzed more accurately. This can reduce the influence of unnecessary background information and complexity of an image by turning a detailed picture into simpler and labelled regions.

These traditional techniques are generally simpler and require fewer computational resources than deep learning models. However, their performance can be affected by lighting conditions, camera quality, and differences between food items, making them less robust in complex real-world environments.

B. Convolutional Neural Networks (CNNs)

Convolutional neural networks are widely used deep learning approaches for image-based food freshness detection. CNNs automatically extract important visual features from food images without requiring manual feature selection. The convolutional layers identify low-level features such as edges, colors, and patterns, while deeper layers learn more complex features such as texture changes, surface damage, and other visible signs of spoilage. The extracted features are then passed to classification layers to predict categories such as fresh, medium, or rotten. CNNs, therefore, provide an effective approach for identifying visual differences between different freshness conditions.

C. Transfer Learning Methods

Transfer learning is basically reusing a model that is already trained on a huge dataset - like Image Net instead of building an entire

model from scratch. It picks up the freshness patterns from an image, way faster and with less data. Common ones people use for this are VGG 16, ResNet, MobileNet and EfficientNet. MobileNet is lighter than others because it cuts down the large calculations and parameters by a lot, without losing an accuracy.

D. Comparison of AI Methods

Different AI-based image detection methods have different advantages and limitations. CNN-based models can provide strong image classification performance but may require considerable computational resources, especially when using deeper architectures. Transfer learning can reduce the amount of training data and time required because the model already contains learned visual features. VGG16 provides a relatively straightforward architecture but has a comparatively large model. ResNet can learn deeper features through residual connections [2]. MobileNet is designed to be lightweight and is therefore more suitable for smartphones and other resource-constrained devices [3]. EfficientNet aims to achieve a balance between accuracy and computational efficiency through compound model scaling [4].

The methods can be compared according to classification accuracy, computational cost, model size, dataset requirements, training time, and suitability for mobile deployment. The most accurate model is not necessarily the most suitable for practical deployment because a larger model may require more memory and processing power. Therefore, selecting an appropriate model involves balancing prediction performance with computational efficiency and deployment requirements.

IV. DATASETS AND EVALUATION METHODS

A. Food Image Datasets

Food freshness datasets should describe the type of food, number of images, class labels, and diversity of image conditions. Images should vary in lighting, backgrounds, camera angles, presentation, and freshness stages. A recent IEEE study on fruits and vegetables compared three-class freshness classification and evaluated VGG16 and YOLOv5, illustrating the value of reporting the class structure and experimental conditions clearly [5].

B. Training and Testing Strategy

The dataset should be divided into training, validation, and testing sets. The training set is used to train the model, the validation set is used to tune parameters and monitor performance, and the testing set is used to evaluate the final model on unseen images. The test data should remain separate from training and model-selection decisions to provide a more reliable estimate of generalization.

C. Evaluation Metrics

The performance of food freshness detection methods can be evaluated using accuracy, F1-score, recall, and precision. These metrics provide a more complete assessment than accuracy alone, particularly when the dataset contains unequal numbers of images in different freshness classes.

D. Confusion Matrix

A confusion matrix can be used to analyze classification performance among the fresh, medium, and rotten classes. It can identify which classes are correctly classified and which classes are frequently confused with one another. For example, confusion between fresh and medium may indicate that gradual visual changes are harder for the model to recognize than clear signs of spoilage.

E. Limitations of Dataset-Based Evaluation

Dataset-based evaluation may not fully represent real-world performance. Images collected under controlled laboratory conditions may differ from real-world images because of variations in lighting, background, camera quality, food presentation, and environmental conditions. Therefore, models should also be evaluated using diverse real-world images. Recent freshness studies also show that reported results depend strongly on the chosen dataset and experimental configuration [5], [6].

V. RECOMMENDED STRATEGIES FOR SAFE USE

A. Confidence-Based Decision Making

The confidence score generated by the model can be used as an indicator of prediction reliability. High-confidence predictions may be presented to the user with appropriate guidance, while low-confidence predictions should be treated with greater caution. Confidence should not be interpreted as a direct measure of food safety.

B. Handling Uncertain Predictions

When the model produces a low-confidence prediction, the system can request another image from the user. The user may be asked to provide an image with better lighting, a closer view, improved framing, or a different angle. This can help reduce errors caused by poor image quality.

C. AI Is Not a Food-Safety Guarantee

AI-based visual freshness detection should not be considered a complete food-safety assessment. Visual analysis cannot completely identify every form of contamination, including pathogens that may not produce visible changes. Therefore, an AI prediction should not be treated as definitive proof that food is safe to consume. Food-safety guidance emphasizes that harmful pathogens may not affect taste, smell, or appearance, so storage and handling information also matters [8].

D. Human Verification

Users should consider additional information along with the AI prediction, including storage temperature, storage duration, expiry or use-by information, and visible or physical signs of deterioration.

E. Safe Disposal Recommendations

When food shows clear signs of severe spoilage or when the system and available evidence indicate substantial uncertainty, the system should provide a cautionary recommendation rather than encouraging consumption. This can help reduce the risk of false reassurance.

VI. MOBILE AND REAL-WORLD DEPLOYMENT

A. Computational Requirements

Deployment on smartphones requires consideration of CPU usage, memory and RAM requirements, processing time, and

energy consumption. Large deep learning models may be unsuitable for devices with limited computational resources.

B. Lightweight Models

Lightweight architectures such as MobileNet and efficient variants of EfficientNet can be considered for mobile deployment because they are designed to improve the trade-off between accuracy and computational cost [3], [4].

C. Model Optimization

Techniques such as quantization, pruning, and model compression can reduce model size and computational requirements, making deployment on mobile and edge devices more practical.

D. Real-World Challenges

Real-world deployment introduces challenges such as different lighting conditions, camera quality, background variation, food diversity, image angles, and environmental conditions. These factors can reduce model performance compared with controlled testing environments. For this reason, evaluation should include images captured under conditions that resemble actual user environments.

VII. PROPOSED SYSTEM / PRACTICAL FRAMEWORK

The proposed system combines food image classification with confidence estimation, safety guidance, and recipe recommendations. The overall workflow can be represented as:

User Image -> Image Preprocessing -> CNN Model -> Freshness Classification -> Confidence Score -> Decision Making -> Safety Guidance -> Recipe Recommendation

The user first captures or uploads an image of food. The image is then preprocessed and passed to the trained CNN model. The model predicts the freshness category and generates a confidence score. Based on the prediction and confidence level, the system provides an appropriate result or requests another image when the prediction is uncertain. For food that appears visually suitable for use, the system can provide recipe recommendations to encourage timely utilization and reduce food waste. The framework is intentionally designed so that recipe recommendation remains separate from a food-safety guarantee.

VIII. CHALLENGES AND LIMITATIONS

Several challenges can affect the reliability of AI-based food freshness detection. These include limited datasets, incorrect or inconsistent image labels, class imbalance, false predictions, poor performance on unseen food types, lighting and image-quality variations, computational limitations, and the inability of image-based models to detect invisible contamination. These limitations should be considered when interpreting model performance and deploying the system in real-world environments.

IX. FUTURE DIRECTIONS

Future research can focus on developing larger and more diverse datasets containing food images from real-world environments. Multimodal systems combining image analysis with temperature, humidity, gas, or other sensor data could provide more comprehensive food-quality assessment. Recent work on sensor-based and mobile food-freshness systems supports the potential value of combining multiple sensing modalities [7], [9]. Explainable AI could help users

understand why a particular freshness prediction was made. Further research can also focus on smartphone deployment, real-time freshness monitoring, lightweight architectures, and improved uncertainty estimation to make these systems more reliable and practical.

X. CONCLUSION

AI and computer vision provide useful tools for food freshness classification because they can learn visual patterns related to color, texture, surface damage, and other visible changes. CNNs and transfer learning offer practical approaches when image datasets are limited, while lightweight architectures such as MobileNet and EfficientNet are relevant to mobile deployment [1]-[4]. Suitable datasets, balanced evaluation metrics, confusion-matrix analysis, and real-world testing are essential for judging model reliability. At the same time, image-based predictions have an important limitation: visual appearance cannot reveal every form of contamination. Therefore, AI-based freshness detection should be used as a decision-support technology rather than a definitive food-safety guarantee. Larger real-world datasets, multimodal sensing, lightweight models, and safety-oriented decision making are promising directions for future development.

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